

AUTONOMOUS

# Programmatic

## SEO Engine

Drop this document into any Lovable project that already has the Content Engine installed. It detects your existing database, generates thousands of SEO pages from your structured data, and indexes them automatically. No writing. No API calls per page. Pure data assembly at scale.

|     |        |          |   |
|-----|--------|----------|---|
| 2–3 | 1,000s | \$0/page | 0 |
|-----|--------|----------|---|

|                 |                 |                 |                    |
|-----------------|-----------------|-----------------|--------------------|
| Lovable credits | Pages generated | Generation cost | Daily intervention |
|-----------------|-----------------|-----------------|--------------------|

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Compatibility: Any Lovable project with the Content Engine installed and Supabase enabled  
Pairs with: Autonomous Content Engine v1.0  
Classification: Confidential

# Table of Contents

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|           |                                                                                 |
|-----------|---------------------------------------------------------------------------------|
| <b>01</b> | <b>How This Engine Works</b><br>The difference between writing and assembling   |
| <b>02</b> | <b>What You Need Before Starting</b><br>Prerequisites and database requirements |
| <b>03</b> | <b>Setup — Two Questions</b><br>Your site type and page volume                  |
| <b>04</b> | <b>The Activation Prompt</b><br>Copy, paste, walk away                          |
| <b>05</b> | <b>Page Type System</b><br>The seven template types and when each fires         |
| <b>06</b> | <b>The Template Engine</b><br>How data becomes pages                            |
| <b>07</b> | <b>The Generation Matrix</b><br>What pages get created from your data           |
| <b>08</b> | <b>URL Structure &amp; Routing</b><br>Clean URLs, canonical tags, no duplicates |
| <b>09</b> | <b>The SEO Layer</b><br>Schema, meta, IndexNow, sitemaps                        |
| <b>10</b> | <b>The Admin Control Panel</b><br>Managing pages, templates, and indexing       |
| <b>11</b> | <b>Content Engine Integration</b><br>How pSEO and written content work together |
| <b>12</b> | <b>Indexing &amp; Discovery Pipeline</b><br>Getting pages found fast            |
| <b>13</b> | <b>Quality Controls</b><br>Thin content protection, deduplication               |
| <b>14</b> | <b>Database Schema</b><br>All tables created by this spec                       |
| <b>15</b> | <b>Multi-Site Scaling</b><br>Replicating across your network                    |
| <b>16</b> | <b>Troubleshooting</b><br>Common issues and fixes                               |

# How This Engine Works

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## The difference between writing and assembling

Your Content Engine writes. Given a keyword, it thinks, drafts, scores, and publishes an original 1,000-word article. It does creative work. That process costs an API call, takes 10–30 seconds, and produces one page. It runs 4 times a day by design.

The Programmatic SEO Engine assembles. It looks at structured data already in your Supabase database — tool names, ratings, pricing, features, pros, cons — and combines that data into page templates. No AI call per page. No writing. No quality gateway. Just data slotted into HTML structure and published to a real URL.

The result is thousands of pages generated in a single run, each targeting a specific long-tail search query that no human writer would ever get around to covering.

### INFO

G2 has over 2 million pages. Capterra has over 1.5 million. Almost none of them were written by a human. They were assembled from user reviews, feature data, and pricing information using exactly this approach.

## What gets generated

The engine looks at your tools table and generates every useful combination:

- **Tool vs Tool pages:** Every pairwise comparison. 50 tools = 2,450 pages.
- **Tool alternatives pages:** 'Best [Tool] alternatives' for every tool in your database.
- **Category index pages:** 'Best tools for [use case]' filtered from your categories.
- **Pricing tier pages:** '[Tool] pricing' with full tier breakdown, updated from your data.
- **Feature comparison pages:** '[Tool] features' pages with structured feature matrices.
- **Use case pages:** 'Best [tool type] for [audience]' from your tags and best\_for fields.
- **Integration pages:** '[Tool A] + [Tool B] integration' if you have integration data.

## How it connects to the Content Engine

Every pSEO page is intentionally thin — useful enough to answer the basic question, not deep enough to replace a full review. Every pSEO page links to the relevant Content Engine article where one exists, and flags the keyword for the Content Engine queue where one does not. The two engines feed each other: pSEO creates surface area and captures long-tail traffic, the Content Engine provides depth and earns authority.

#### INFO

Think of it this way: the pSEO engine catches people searching 'Bubble vs Adalo'. It gives them a quick comparison and sends them to your full Bubble review and full Adalo review. Both of those were written by your Content Engine. The visitor converts on one of those pages.

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## SECTION 02

# What You Need Before Starting

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This engine requires your Supabase database to have data in it before it can generate pages. An empty database generates nothing. The more data you have, the more pages get created.

## Required: Content Engine already installed

The Content Engine spec must already be active in this Lovable project. The pSEO engine uses the same Supabase instance, the same articles table, the same site\_settings table, and the same affiliate\_links table. It adds new tables on top — it does not replace anything.

## Required: Data in your tools table

The engine reads from a tools table. If your project was built from the Site 1 spec (No-Code Platform Reviews), this table already exists and may have data. If not, the activation prompt creates the table and seeds it with 15 starter tools. You can add more tools via the admin panel at any time.

### Minimum data for useful page generation:

| Requirement      | Minimum                                                             | Notes                                                                                  |
|------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| tools table rows | Minimum 10                                                          | Pages generated scale with tool count. 10 tools = ~200 pages. 50 tools = ~5,000 pages. |
| Fields populated | name, slug, category, rating_overall, pros[], cons[], pricing_tiers | These fields power the comparison and review templates. Empty fields = thin pages.     |
| affiliate_links  | At least 1 row                                                      | pSEO pages include affiliate CTAs pulled from this table.                              |
| site_settings    | Site identity block                                                 | Used in page meta descriptions and footer context.                                     |

## Optional but recommended

- **Google Search Console token** — enables the engine to prioritise page generation based on which comparisons are already getting impressions.
- **Comparisons table** — if you have manually written any comparison articles via the Content Engine, the pSEO engine detects them and links to them instead of generating a thin version.
- **20+ tools** — under 10 tools, pSEO adds limited value over the Content Engine alone. The exponential growth in pages kicks in at 20+ tools.

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## SECTION 03

# Setup — Two Questions

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Before pasting the activation prompt, answer these two questions. Your answers control what gets generated and at what volume.

## Question 1 — Your site type

The site type tells the engine which page templates to prioritise and which URL patterns to use. Choose the one that best matches your anchor site.

| Site type    | Use when                              | Extra templates unlocked                                               |
|--------------|---------------------------------------|------------------------------------------------------------------------|
| tool-reviews | You review specific software tools    | Comparison, alternatives, pricing, features pages. Default for Site 1. |
| automation   | You cover workflow / automation tools | Integration pages added. Tool + platform combination pages.            |
| ecommerce    | You cover ecommerce platforms         | Fee calculator data pages, platform-for-use-case pages.                |
| saas         | You cover SaaS building tools         | Stack combination pages, stage-based recommendation pages.             |
| directory    | You run a business/service directory  | Location + category combination pages at scale.                        |
| generic      | None of the above                     | Comparison and alternatives pages only.                                |

## Question 2 — Your page volume target

This controls how aggressively the engine generates pages on first run and how many new pages it adds each week as you add tools.

| Volume       | What it does                         | Best for                                                   | Tools cap     |
|--------------|--------------------------------------|------------------------------------------------------------|---------------|
| conservative | Generate top 200 pages only          | New sites, <20 tools, want to avoid thin content penalties | ~50 tools max |
| standard     | Generate all valid combinations      | Established sites, 20–100 tools, good data quality         | Unlimited     |
| aggressive   | Generate all combinations + variants | High-authority sites, 50+ tools, excellent data quality    | Unlimited     |

## INFO

Start with 'standard'. After 60 days, check Google Search Console for which pSEO pages are getting impressions. If quality is good, switch to 'aggressive' to fill the gaps.

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## SECTION 04

# The Activation Prompt

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Paste this entire block into Lovable's chat input. Replace the two [BRACKETED] sections with your answers from Section 03. Do not split it into multiple messages.

### CRITICAL

This prompt uses approximately 2–3 Lovable credits. It detects your existing Content Engine installation and adds only what is missing. It will not overwrite any existing tables, routes, or components.

## ACTIVATION PROMPT — COPY THIS ENTIRE BLOCK INTO LOVABLE

```
Build and activate a complete Programmatic SEO Engine for this Lovable project.
Before building anything, audit the existing codebase for: (1) Content Engine
installation, (2) tools table in Supabase, (3) existing pseo_pages table,
(4) existing pSEO routes, (5) sitemap generation system.
For each component found working: skip it and log "found existing -- skipped".
For each component missing or incomplete: build from scratch.
```

```
SITE TYPE: [YOUR SITE TYPE FROM SECTION 03]
```

```
PAGE VOLUME: [YOUR VOLUME TARGET: conservative / standard / aggressive]
```

```
BUILD THE FOLLOWING COMPLETE SYSTEM:
```

### 1. TOOLS TABLE

```
If tools table does not exist: create it with fields id (uuid), name (text),
slug (text unique), tagline (text), logo_url (text), website_url (text),
affiliate_url (text), category (text[]), rating_overall (decimal),
rating_ease (decimal), rating_features (decimal), rating_pricing (decimal),
verdict (text), review_body (text), pros (text[]), cons (text[]),
pricing_tiers (jsonb), best_for (text[]), featured (bool), sponsored (bool),
created_at (timestamp), updated_at (timestamp).
```

```
If table exists: add any missing columns only. Seed with 15 starter no-code
tools if row count is zero.
```

### 2. PSEO\_PAGES TABLE

```
Create if not exists: id (uuid), page_type (text), slug (text unique),
```



title (text), meta\_description (text), h1 (text), tool\_id\_primary (uuid), tool\_id\_secondary (uuid nullable), category (text nullable), body\_html (text), status (text: draft/published/suppressed), quality\_score (integer), generated\_at (timestamp), indexed\_at (timestamp nullable), search\_impressions (integer default 0), search\_clicks (integer default 0), content\_engine\_article\_id (uuid nullable).

### 3. PAGE GENERATION ENGINE

On activation: generate all valid page combinations based on site type and volume setting. Page types to generate:

TYPE A -- tool\_vs\_tool: for every pair of tools where both have rating\_overall and at least 3 pros and 3 cons populated. Slug: /compare/[tool-a]-vs-[tool-b]. Title: "[Tool A] vs [Tool B]: Which is better in [current year]?"  
Generate winner verdict from rating comparison. Include pricing table, pros/cons side-by-side, feature matrix from pricing\_tiers data, final verdict with affiliate CTAs for both tools. Link to Content Engine article if one exists for this comparison keyword.

TYPE B -- alternatives: for every tool with rating\_overall >= 3.5 and at least 5 other tools in same category. Slug: /alternatives/[tool-name]. Title: "Best [Tool Name] Alternatives in [current year]".  
List top 5 alternatives ranked by rating, each with 2-sentence description from tagline + verdict fields, pros vs original tool, affiliate CTA.

TYPE C -- category\_best: for every unique value in category[] field where 5+ tools share that category. Slug: /best/[category-slug].  
Title: "Best [Category] Tools in [current year]".  
Ranked list of tools in category, sorted by rating\_overall DESC.  
Each entry: logo, name, one-line verdict, rating, pricing start point, affiliate CTA.

TYPE D -- pricing: for every tool where pricing\_tiers jsonb has 2+ tiers. Slug: /pricing/[tool-name].  
Title: "[Tool Name] Pricing: All Plans Explained [current year]".  
Full pricing tier breakdown table, free plan indicator, money-back guarantee if in data, comparison to category average price, affiliate CTA.

TYPE E -- features: for every tool with rating\_features populated and 3+ pros. Slug: /features/[tool-name].  
Title: "[Tool Name] Features: Complete Breakdown".  
Feature list from pros[], key capabilities from pricing\_tiers, best\_for[] use cases as tags, limitations from cons[], verdict from verdict field.

TYPE F -- use\_case (site\_type = tool-reviews or saas only): for every unique value in best\_for[] field where 3+ tools share that use case. Slug: /for/[use-case-slug].  
Title: "Best No-Code Tools for [Use Case]".  
Ranked tool list filtered by best\_for tag, each with affiliate CTA.

TYPE G -- integrations (site\_type = automation only): for every tool pair that appears together in a keyword context. Slug: /integrate/[tool-a]-[tool-b]. Title: "How to connect [Tool A] and [Tool B] without code". Generated from tool descriptions and category data.

#### 4. QUALITY FILTER

Before publishing any page, check:

- Primary tool must have: name, slug, rating\_overall, at least 2 pros, at least 2 cons, affiliate\_url
- tool\_vs\_tool pages: both tools must pass above check
- If quality check fails: set status = draft, log reason to pseo\_quality\_log
- Minimum quality score 60/100 to publish.  $\text{Score} = (\text{fields\_populated} / \text{total\_fields} * 100)$

#### 5. THIN CONTENT PROTECTION

Never generate a page if: word count of assembled content < 300 words.

Never generate tool\_vs\_tool if both tools are in different primary categories (e.g. website builder vs database tool -- not a useful comparison).

Suppress duplicate semantic intent: if Content Engine has published an article with >80% keyword overlap, link to that article instead of generating a pSEO page.

#### 6. ROUTING

Create dynamic routes for all page types:

/compare/[slug], /alternatives/[slug], /best/[slug],  
/pricing/[slug], /features/[slug], /for/[slug], /integrate/[slug].

All routes server-side rendered. Each page loads data from pseo\_pages table.

404 if status != published.

## 7. SEO LAYER

Every pSEO page must have:

- Unique meta title (from title field, max 60 chars)
- Meta description (from meta\_description field, max 155 chars)
- Canonical URL (self-referencing, no trailing slash)
- Open Graph tags (og:title, og:description, og:url, og:type=article)
- JSON-LD schema appropriate to page type:
  - \* tool\_vs\_tool: ItemList schema with both tools
  - \* alternatives: ItemList schema
  - \* category\_best: ItemList schema
  - \* pricing: SoftwareApplication schema with offers
  - \* features: SoftwareApplication schema
- hreflang if multiple language versions exist
- Last-modified header set to generated\_at or updated\_at
- Sitemap entry added on publish

## 8. INTERNAL LINKING

Every pSEO page must include:

- Link to the primary tool's full review (Content Engine article if exists, otherwise /tools/[slug])
- For tool\_vs\_tool: link to each tool's alternatives page
- For alternatives: link to tool\_vs\_tool page for top 2 alternatives
- For category\_best: link to top 3 tools' individual pricing pages

- Cross-link to Content Engine article where keyword overlap > 60%
- Max 8 internal links per page

#### 9. AFFILIATE INTEGRATION

Pull affiliate URLs from affiliate\_links table by tool\_name match.

Fallback to tools.affiliate\_url field.

Use default\_fallback from affiliate\_links if no match found.

CTA button text: "Try [Tool Name] Free" if free plan exists, else "View [Tool Name]".

Max 2 affiliate CTAs per pSEO page (primary tool + one comparison/alternative).

#### 10. INDEXING PIPELINE

On each new page published:

- Add URL to sitemap.xml (auto-regenerate)
- Submit to IndexNow API using INDEXNOW\_KEY environment variable
- Log to pseo\_indexing\_log: url, submitted\_at, response\_code
- Batch submissions: max 100 URLs per IndexNow call

#### 11. WEEKLY REFRESH AGENT

Every Monday 08:00 UTC:

- Check tools table for updated\_at > last page generation for that tool
- Regenerate all pSEO pages for updated tools
- Update pricing pages if pricing\_tiers changed
- Add new pages for any tools added since last run
- Log run to pseo\_refresh\_log

## 12. ADMIN PANEL EXTENSION

Add a "PROGRAMMATIC SEO" section to the existing /content-admin panel:

- Page count by type (table: page\_type, count, published, draft, suppressed)
- Generation log: last run timestamp, pages created, pages updated, errors
- Quality failures: list of draft pages with failure reason, fix suggestions
- Manual regenerate button: rebuilds all pages for a selected tool
- Suppress page button: marks a page as suppressed (hides from site + sitemap)
- Indexing status: pages submitted to IndexNow, awaiting indexing, confirmed indexed
- Top performing pSEO pages: sorted by impressions (if GSC token available)

## 13. SEARCH CONSOLE INTEGRATION (optional)

If GOOGLE\_SEARCH\_CONSOLE\_TOKEN is set:

- Weekly: pull impression and click data for all /compare/, /alternatives/, /best/, /pricing/, /features/ URLs
- Update pseo\_pages.search\_impressions and search\_clicks fields
- Flag pages with 0 impressions after 90 days for regeneration with stronger angle
- Flag pages with impressions > 100 but CTR < 1% for title/meta rewrite
- Surface top 10 performing pSEO pages in admin panel

## 14. DATABASE TABLES

Create if not existing: pseo\_pages, pseo\_quality\_log, pseo\_indexing\_log, pseo\_refresh\_log. Add pseo\_page\_count (integer) to network\_reports table for multi-site dashboard.

AFTER BUILDING: Provide a build report listing:

- Total pages generated by type
- Pages published vs draft (quality failures)
- Any tools with insufficient data for page generation
- Admin panel URL for the new pSEO section
- Environment variables needed: INDEXNOW\_KEY (get free at [indexnow.org](https://indexnow.org))

## Follow-up prompts if needed

Use these only if the build report flags specific issues:

### If pages are generating but not routing correctly:

The pSEO routes are not resolving. Debug the dynamic routing for /compare/, /alternatives/, /best/, /pricing/, /features/, /for/ and /integrate/ paths. Confirm each route reads from the pseo\_pages table and returns 404 for status != published. Test /compare/[first-slug] and show me the output.

### If quality failures are blocking most pages:

Most pSEO pages are in draft status due to quality failures. Show me the pseo\_quality\_log table with the top 5 failure reasons. Then check the tools table and tell me which fields are most commonly empty. I want to fix the data, not lower the quality threshold.

### **If IndexNow submissions are failing:**

IndexNow submissions are returning errors. Verify the INDEXNOW\_KEY environment variable is set and the submission format matches the IndexNow API spec. Show me the last 5 rows from pseo\_indexing\_log with their response codes.

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## SECTION 05

# Page Type System

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## The seven template types and when each fires

Each page type targets a different search intent and requires different data. The engine only generates a page type if it has sufficient data to make it useful.

**Typ  
e A**

**tool\_vs\_tool**

**URL:** /compare/[tool-a]-vs-[tool-b]

Comparison intent. Highest buyer intent of all page types. Visitor is actively choosing between two tools.

*Requires: Both tools need: rating, 3+ pros, 3+ cons, pricing\_tiers, affiliate\_url*

**Typ  
e B**

**alternatives**

**URL:** /alternatives/[tool-name]

Alternative-seeking intent. Visitor is unhappy with current tool or researching options before committing.

*Requires: Primary tool needs: rating >= 3.5. 5+ other tools in same category.*

**Typ  
e C**

**category\_best**

**URL:** /best/[category]

Discovery intent. Visitor knows the category, wants to see the field. High volume, lower conversion than comparison.

*Requires: 5+ tools sharing a category tag.*

**Typ  
e D**

**pricing**

**URL:** /pricing/[tool-name]

Pricing intent. Visitor is close to purchasing, comparing costs. High affiliate conversion rate.



*Requires: Tool needs: 2+ pricing tiers in pricing\_tiers jsonb field.*

**Typ  
e E**

**features**

**URL:** /features/[tool-name]

Research intent. Visitor evaluating capabilities before trialling. Good for tools with complex feature sets.

*Requires: Tool needs: rating\_features, 3+ pros, verdict field populated.*

**Typ  
e F**

**use\_case**

**URL:** /for/[use-case]

Job-to-be-done intent. Visitor knows their goal, wants the best tool. Very high conversion when matched correctly.

*Requires: 3+ tools sharing a best\_for[] tag value.*

**Typ  
e G**

**integration**

**URL:** /integrate/[tool-a]-[tool-b]

Integration intent. Visitor is building a stack and checking compatibility. Automation sites only.

*Requires: Automation site type selected. Both tools in database.*

# The Template Engine

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## How data becomes pages

Every page type has a fixed HTML template with data slots. When the engine runs, it queries Supabase for the relevant tools, fills the slots, assembles the HTML, calculates a quality score, and writes the result to the `pseo_pages` table. No AI. No API call. No latency beyond a database query.

### The `tool_vs_tool` template structure

1. Hero section — Tool A logo, name, overall rating vs Tool B logo, name, overall rating. Winner badge on higher-rated tool.
2. Quick verdict — One sentence from each tool's verdict field. If verdict is empty, generated from rating + `pros[0]`.
3. Pricing comparison table — `pricing_tiers` jsonb for both tools, side by side. Lowest tier highlighted.
4. Pros and cons comparison — `pros[]` and `cons[]` arrays rendered as two-column table. Up to 5 items per tool.
5. Feature matrix — key features from `pricing_tiers`, checkmark/cross per tool.
6. Best for section — `best_for[]` tags for each tool rendered as chips.
7. Verdict and CTA — rating-based winner declaration, affiliate CTA buttons for both tools.
8. Related links — links to each tool's alternatives page, pricing page, and Content Engine article if exists.

## Data freshness

Every time a tool's record is updated in Supabase — pricing changes, new pros added, rating updated — the weekly refresh agent regenerates all pSEO pages that reference that tool. The `updated_at` timestamp on the `tools` table is the trigger. Pages always reflect current database state.

### INFO

This is a significant advantage over static content. When Bubble raises its prices, you update one row in Supabase. Every Bubble vs X comparison page, every Bubble pricing page, and every Bubble alternatives page automatically reflects the new pricing on the next weekly refresh.

# The Generation Matrix

## What pages get created from your data

Use this table to estimate how many pages the engine will generate at different tool counts. Numbers assume standard volume setting and full data quality.

| Tool count | Categories    | vs pages | Alt pages | Cat pages | Pricing | Features | Total   |
|------------|---------------|----------|-----------|-----------|---------|----------|---------|
| 10 tools   | 3 categories  | ~45      | ~10       | ~3        | ~10     | ~10      | ~78     |
| 25 tools   | 6 categories  | ~300     | ~25       | ~6        | ~25     | ~25      | ~381    |
| 50 tools   | 10 categories | ~1,225   | ~50       | ~10       | ~50     | ~50      | ~1,385  |
| 100 tools  | 15 categories | ~4,950   | ~100      | ~15       | ~100    | ~100     | ~5,265  |
| 200 tools  | 20 categories | ~19,900  | ~200      | ~20       | ~200    | ~200     | ~20,520 |

### INFO

The tool\_vs\_tool count is  $n*(n-1)/2$  — it grows exponentially. 100 tools gives you nearly 5,000 comparison pages. This is why data quality matters more than tool count: 200 well-documented tools beats 500 half-empty records every time.

## Priority ordering

The engine does not generate all pages at once in random order. It prioritises pages likely to rank and convert:

1. **tool\_vs\_tool for featured tools** — tools marked featured=true generate comparisons first.
2. **High-rated tools** — tools with rating\_overall  $\geq 4.0$  generate all page types before lower-rated tools.
3. **Sponsored tools** — tools with sponsored=true and active sponsored\_until date generate all page types on the next run.
4. **Most-linked tools** — tools referenced most often in Content Engine articles (via affiliate\_clicks table) generate first.
5. **Everything else** — remaining tools in alphabetical order.

# URL Structure & Routing

## Clean URLs, canonical tags, no duplicates

URL structure matters for SEO. The engine uses a consistent, predictable pattern across all page types.

| Page type     | URL pattern                             | Notes                                                                                       |
|---------------|-----------------------------------------|---------------------------------------------------------------------------------------------|
| tool_vs_tool  | /compare/[tool-a-slug]-vs-[tool-b-slug] | Tool slugs alphabetically ordered to prevent /compare/a-vs-b and /compare/b-vs-a duplicates |
| alternatives  | /alternatives/[tool-slug]               | Singular tool slug only                                                                     |
| category_best | /best/[category-slug]                   | Category slug from tools.category[] value                                                   |
| pricing       | /pricing/[tool-slug]                    | Tool slug only                                                                              |
| features      | /features/[tool-slug]                   | Tool slug only                                                                              |
| use_case      | /for/[use-case-slug]                    | Slugified best_for[] value                                                                  |
| integration   | /integrate/[tool-a-slug]-[tool-b-slug]  | Alphabetically ordered                                                                      |

## Duplicate prevention

- tool\_vs\_tool slugs always sort tool names alphabetically — bubble-vs-webflow not webflow-vs-bubble.
- 301 redirect created automatically for the reverse URL pattern.
- If a Content Engine article exists at /blog/[similar-slug], canonical points to the article, not the pSEO page.
- Suppressed pages return 410 Gone, not 404, to signal intentional removal to Google.

# The SEO Layer

## Schema, meta, IndexNow, sitemaps

Every pSEO page is fully optimised for search on publish. None of this requires manual intervention.

### Meta generation rules

| Page type     | Title pattern                                            | Meta description pattern                                                                                           |
|---------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| tool_vs_tool  | [Tool A] vs [Tool B]: Which is better in [year]?         | [Tool A] and [Tool B] compared: pricing, features, pros and cons. Find out which is right for your use case.       |
| alternatives  | Best [Tool] Alternatives in [year]: [N] Options Compared | Looking for [Tool] alternatives? We compared the top [N] options on pricing, features and ease of use.             |
| category_best | Best [Category] Tools in [year]: Ranked and Reviewed     | [N] [category] tools ranked by real ratings. Compare pricing, features and find the right tool for your needs.     |
| pricing       | [Tool] Pricing [year]: All Plans and What You Get        | [Tool] offers [N] pricing plans from \$[X]/month. Here is exactly what each plan includes and which one to choose. |
| features      | [Tool] Features: Complete List and What They Do          | Everything [Tool] can do — a complete breakdown of features, use cases, and limitations before you sign up.        |

### JSON-LD schema per page type

- **tool\_vs\_tool:** ItemList with two ListItem entries, each a SoftwareApplication.
- **alternatives:** ItemList with up to 5 SoftwareApplication entries.
- **category\_best:** ItemList with all tools in category.
- **pricing:** SoftwareApplication with Offer array for each pricing tier.
- **features:** SoftwareApplication with full feature description.

## IndexNow setup

IndexNow allows instant URL submission to Bing and a growing number of search engines (including Yandex). Google has indicated support is coming. Get a free key at [indexnow.org](https://indexnow.org) — it takes 60 seconds.

1. Go to [indexnow.org](https://indexnow.org) and generate a key.
2. Add it as INDEXNOW\_KEY in your Lovable environment variables.
3. The engine automatically submits every new page URL on publish.
4. Batch submissions of up to 100 URLs are sent per IndexNow call.

5. Submission logs are viewable in the admin panel under Programmatic SEO > Indexing Status.

## **Sitemap management**

The engine maintains a separate sitemap-pseo.xml alongside the main sitemap.xml from the Content Engine. Both are referenced in robots.txt. The pSEO sitemap includes all published pages with lastmod set to generated\_at or the last refresh date. Suppressed pages are excluded.

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## SECTION 10

# The Admin Control Panel

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The Programmatic SEO section is added to your existing /content-admin panel. It does not create a new route — it extends what is already there.

### Overview tab

Total pages published, by type, as a breakdown table.

Pages in draft (quality failures) with a link to the quality failure log.

Pages suppressed manually.

Last generation run timestamp and next scheduled refresh.

IndexNow submission queue — pages awaiting submission, submitted, confirmed.

### Tools data quality tab

Table of all tools with a completeness score (0–100%) based on fields populated.

Fields missing highlighted in amber — click to edit the tool directly.

'Generate pages now' button per tool — immediately regenerates all pSEO pages for that tool.

Bulk import for tools — paste a CSV of tool data to populate multiple tools at once.

### Performance tab (requires Google Search Console token)

Top 20 pSEO pages by impressions over the last 28 days.

Pages with high impressions but low CTR — flagged for title/meta rewrite.

Pages with zero impressions after 90 days — flagged for regeneration or suppression.

Click-through rate trend chart for pSEO pages vs Content Engine articles.

### Page management tab

Full list of all pseo\_pages with status, type, quality score, impressions.

Filter by type, status, quality score range.

Suppress button — removes page from site and sitemap, sends 410 to crawlers.

Regenerate button — rebuilds page from current Supabase data.

View live page link for all published pages.

# Content Engine Integration

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## How pSEO and written content work together

The two engines are designed to complement, not compete. The integration has three parts:

### Part 1 — Deduplication

Before generating any pSEO page, the engine checks the articles table for Content Engine articles with keyword overlap. If overlap > 60%, the pSEO page is suppressed and the Content Engine article is linked from the pSEO page's canonical URL. The pSEO page still exists in the database — it is just not published. This prevents Google from seeing two pages competing for the same query.

### Part 2 — Keyword handoff

Every pSEO page that gets more than 50 impressions per month (from Google Search Console data) and does not have a corresponding Content Engine article is automatically added to the Content Engine keyword queue with high priority. The signal is clear: Google likes this topic. Write a proper article about it.

### Part 3 — Link flow

Content Engine articles link to relevant pSEO pages. pSEO pages link to relevant Content Engine articles. Both link to tool review pages. The internal link structure is intentional: pSEO pages have high volume and funnel link equity to your best Content Engine content.

#### INFO

The ideal state: a visitor lands on your 'Bubble vs Webflow' pSEO page from a generic search. They get a quick comparison. They click through to your full Bubble review — a 2,000-word Content Engine article. They sign up via your affiliate link. pSEO captured the traffic. Content Engine converted it.



# Indexing & Discovery Pipeline

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## Getting pages found fast

Generating thousands of pages means nothing if they are not indexed. The pipeline moves pages from generated to indexed as efficiently as possible.

### Phase 1 — IndexNow (day 1)

On every publish batch, the engine submits all new URLs to IndexNow. This gets pages into Bing and other participating search engines within 24–48 hours. It also signals to Google that new content exists, even before direct Google submission.

### Phase 2 — Google Search Console API (ongoing)

If `GOOGLE_SEARCH_CONSOLE_TOKEN` is set, the engine submits URLs directly to the Google Search Console URL Inspection API for priority pages (featured tools, high-priority comparisons). This is rate-limited to 200 URLs/day by Google.

### Phase 3 — Sitemap ping (weekly)

Every Monday after the weekly refresh, the engine pings Google's sitemap endpoint with the updated `sitemap-pseo.xml` URL. This is the slowest but most reliable method for bulk page discovery.

### Phase 4 — Internal links (immediate)

New pSEO pages are linked from category index pages and from related Content Engine articles. When Googlebot crawls those existing pages, it discovers the new URLs naturally. This is often faster than any direct submission method.

#### INFO

Do not submit all 5,000 pages to Google Search Console manually. Google's crawler will find them via the sitemap and internal links. Manual submission is only worth it for your highest-priority 50 pages.

# Quality Controls

## Thin content protection and deduplication

Thin content is the biggest risk with programmatic SEO. Google's Helpful Content system specifically targets sites that generate large volumes of low-value pages. These controls exist to prevent that.

### Quality score calculation

| Dimension         | Points | What is checked                                                                                                                                           |
|-------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fields populated  | 30 pts | 10 fields checked: name, slug, tagline, rating_overall, pros (min 3), cons (min 3), pricing_tiers (min 2 tiers), verdict, affiliate_url, best_for (min 2) |
| Data uniqueness   | 25 pts | Pros and cons must be distinct — not the same text recycled. Pricing tiers must have different feature lists.                                             |
| Word count        | 25 pts | Assembled page body must be 300+ words to score any points here. 500+ words scores full marks.                                                            |
| No data conflicts | 20 pts | Rating fields must be consistent. Pricing data must be parseable. No null values in required template slots.                                              |

Pages scoring below 60 are set to draft status. They appear in the admin panel quality failures list with specific fields flagged. Fix the data in Supabase, hit regenerate, and they publish automatically.

### Hard suppression rules

- The engine will never publish a page if:
- Tool has no affiliate\_url and no entry in affiliate\_links table.
  - tool\_vs\_tool where both tools are in completely different primary categories.
  - Page body would contain fewer than 300 words after assembly.
  - Duplicate slug already exists in pseo\_pages or articles table.
  - Primary tool has been marked active=false.

CRITICAL

If Google sends a Manual Action for thin content, immediately run the quality audit in the admin panel. Set all pages scoring below 70 to suppressed. This typically resolves thin content manual actions within 2–4 weeks.

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## SECTION 14

# Database Schema

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### All tables created or extended by this spec

#### pseo\_pages

| Field                     | Type      | Description                                                                               |
|---------------------------|-----------|-------------------------------------------------------------------------------------------|
| id                        | uuid      | Primary key                                                                               |
| page_type                 | text      | tool_vs_tool / alternatives / category_best / pricing / features / use_case / integration |
| slug                      | text      | Unique. Full URL path e.g. /compare/bubble-vs-webflow                                     |
| title                     | text      | SEO title — max 60 chars                                                                  |
| meta_description          | text      | Meta description — max 155 chars                                                          |
| h1                        | text      | Page H1 heading                                                                           |
| tool_id_primary           | uuid      | FK to tools table                                                                         |
| tool_id_secondary         | uuid      | FK to tools table. Null for single-tool pages                                             |
| category                  | text      | For category_best and use_case pages                                                      |
| body_html                 | text      | Assembled page HTML                                                                       |
| status                    | text      | draft / published / suppressed                                                            |
| quality_score             | integer   | 0–100. Must be >= 60 to publish                                                           |
| generated_at              | timestamp | When page was generated                                                                   |
| indexed_at                | timestamp | When IndexNow submission was confirmed                                                    |
| search_impressions        | integer   | From Google Search Console. Updated weekly                                                |
| search_clicks             | integer   | From Google Search Console. Updated weekly                                                |
| content_engine_article_id | uuid      | FK to articles table if a related CE article exists                                       |
| updated_at                | timestamp | Last regeneration time                                                                    |

#### pseo\_quality\_log

| Field         | Type      | Description                      |
|---------------|-----------|----------------------------------|
| id            | uuid      | Primary key                      |
| page_slug     | text      | FK to pseo_pages.slug            |
| quality_score | integer   | Score at time of check           |
| failures      | jsonb     | Array of {field, reason} objects |
| checked_at    | timestamp | When quality check ran           |

### pseo\_indexing\_log

| Field         | Type      | Description                     |
|---------------|-----------|---------------------------------|
| id            | uuid      | Primary key                     |
| url           | text      | Full URL submitted              |
| submitted_at  | timestamp | When submitted to IndexNow      |
| response_code | integer   | HTTP response from IndexNow API |
| engine        | text      | indexnow / gsc / sitemap        |

### pseo\_refresh\_log

| Field         | Type      | Description                       |
|---------------|-----------|-----------------------------------|
| id            | uuid      | Primary key                       |
| run_at        | timestamp | When refresh ran                  |
| pages_updated | integer   | Pages regenerated                 |
| pages_created | integer   | New pages created                 |
| tools_updated | integer   | Tools that triggered regeneration |
| errors        | integer   | Pages that failed to regenerate   |

### tools table — extended fields

The following columns are added to the existing tools table if not present:

- **pseo\_enabled** (boolean, default true) — set to false to exclude a tool from all pSEO generation.
- **pseo\_last\_generated** (timestamp) — when pSEO pages were last generated for this tool.
- **pseo\_page\_count** (integer) — number of published pSEO pages featuring this tool.

# Multi-Site Scaling

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## Replicating across your network

The pSEO engine follows the same replication model as the Content Engine. Each site in your network gets its own instance, its own tools table, and its own pseo\_pages. The master dashboard aggregates stats across all sites.

### The 5-minute replication process

1. Open the target Lovable project (must already have Content Engine installed).
2. Add INDEXNOW\_KEY to environment variables (same key works across sites).
3. Upload this document to Lovable's chat.
4. Select your site type and volume target for the new site.
5. Paste the activation prompt with the correct answers.
6. Populate the tools table with site-specific tools via the admin panel.

### Cross-site considerations

- Each site has completely separate tools and pseo\_pages tables — no data sharing.
- The same tool can exist on multiple sites with different review data.
- network\_reports table is extended with pseo\_page\_count so the master dashboard shows pSEO performance per site.
- Cross-site linking between pSEO pages is handled by the Interlink Agent from the site build spec — the pSEO engine does not manage cross-site links independently.

#### INFO

The 10-site network in your site build spec would generate over 50,000 pSEO pages combined once each site reaches 50+ tools. This is the surface area that makes a content network genuinely hard to compete with.

# Troubleshooting

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## Common issues and fixes

### Pages are generating but showing blank sections

The most common cause is empty fields in the tools table. Open the admin panel > Tools data quality tab. Any tool with a completeness score below 70% will produce pages with empty sections. Fill in the missing fields and hit Regenerate for that tool.

### Most pages are stuck in draft

Check the pseo\_quality\_log in the admin panel. The failure reasons will tell you which fields are causing the most failures. Common issues: pros[] has fewer than 3 items, pricing\_tiers is not valid JSON, affiliate\_url is missing.

### tool\_vs\_tool pages are comparing unrelated tools

The engine compares tools in the same category. If your tools do not have category[] populated consistently (e.g. some say 'app-builder' and others say 'app builder'), the engine cannot match them correctly. Standardise category values across all tools and re-run generation.

### IndexNow submissions are returning 400 errors

Verify INDEXNOW\_KEY matches exactly what is in your indexnow.org account. Also verify that the key verification file exists at your domain root: /[your-key].txt must be accessible. The activation prompt creates this file — check it exists at your Lovable project's public URL.

### Google is not indexing pSEO pages after 90 days

First check: are the pages linked from anywhere? Unlinked pages are rarely crawled. Add your site's category index pages (e.g. /best/app-builders) to your homepage navigation — this gives Google a crawl path to all pSEO pages in that category.

Second check: is sitemap-pseo.xml returning valid XML? Validate it at [sitemap.xml.com](https://www.sitemap.xml.com). A malformed sitemap silently prevents indexing.

Third check: are any pSEO page URLs returning errors? A 404 or 500 on even one URL can slow sitemap processing for the entire file.

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## Environment Variables Checklist

| Variable                    | Where to get it                       | Required?    |
|-----------------------------|---------------------------------------|--------------|
| INDEXNOW_KEY                | indexnow.org — free, instant          | YES          |
| GOOGLE_SEARCH_CONSOLE_TOKEN | Google Search Console API credentials | Optional     |
| GEMINI_API_KEY              | Already set by Content Engine         | Already done |
| RESEND_API_KEY              | Already set by Content Engine         | Already done |
| SITE_ID                     | Already set by Content Engine         | Already done |

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**Drop this document onto any Lovable project running the Content Engine. Answer two questions. Paste one prompt. Walk away.**

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